UCE702: ADVANCED CONSTRUCTION TECHNOLOGY

	L	T	P	Cr
	3	**1**	**0**	**3.5**

Course Objective: The basic objective of the course is to expose the students to the latest and advanced construction techniques, which is fast and reliable. Students will be introduced to building information modeling software, which helps in management of construction site including automation process at site.

Pre-Cast and Prefabricated Structures

Introduction: Need for prefabrication and pre-cast, Materials, Modular coordination, Standardization, Systems, Production, Transportation, Erection.

Behaviour of Structural Components: Large panel constructions, Construction of roof and floor slabs, Wall panels, Columns and Shear walls.

Design Principles: Disuniting of structures; Design of cross section based on efficiency of material used; Problems in design because of joint flexibility and allowance for joint deformation. Joint in structural members, Joints for different structural connections, Dimensions and detailing of expansion joints.

Bridge Construction Methods
Construction procedure: segmental construction; Precast vs cast-in-place; Cantilever construction; Incremental launching method; Accelerated bridge construction. Factors affecting the selection of bridge construction methods.

Automation in Construction
3D Printing: Introduction, Principle, methods, materials, Advantages and disadvantages. Applications in construction industry: Pre-fabrications, Development of building elements, erection of contours, topologies and urban planning details, printing a building.

Robotics in Construction: Objective; Types of robots used in construction. Application: Demolition; Surveying; Paving; concrete finishing; Welding; Brick laying; Risk and Cost Benefit Analysis

Course Learning Objectives (CLO)

Upon the completion of this course, the students will be able to:

1. Design the prefabricated elements and also have the knowledge of the construction methods with prefabricated elements
2. Relate the challenges associated with different construction methods of bridges
3. Understand the 3D Printing process in construction
4. Understand the different automation process in construction

Text Books:
1. CBRI, Building materials and components, India, 1990
2. Gerostiza C.Z., Hendrikson C. and Rehat D.R., “Knowledge based process planning for construction and manufacturing”, Academic Press Inc., 1994

Reference Books:
1. Met Koncz T., “Manual of precast concrete construction”, Vol. I, II and III, Bauverlag, GMBH, 1976.
2. “Structural design manual”, Precast concrete connection details, Society for the studies in the use of precast concrete, Netherland BetorVerlag, 2009

Evaluation Scheme:

Sr. No.	**Evaluation Elements**	**Weights (%)**
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25